

COSC 39: Theory of Computation

Credit Statement: Talked to Sair Shaikh'26, Nand Patel'27, Kiran Jones'27, and Carson Bates'27.

Problem 1. A **dominating set** in graph G is set of vertices S so that every vertex is in S or a neighbor of S .

DOMSET

- **Input:** An undirected graph G , a parameter q
- **Output:** Does G have a dominating set of size at most q ?

We can also separate the parameter q out from the definition of DOMSET:

q -DOMSET

- **Input:** An undirected graph G
- **Output:** Does G have a dominating set of size at most q ?

Prove that q -DOMSET cannot be solved in $O(n^{q-\delta})$ time for any $\delta > 0$ under SETH, by reducing an arbitrary n -variable sparse k -SAT instance to a q -DOMSET instance with $O(q2^{\frac{n}{q}} + m)$ nodes

Solution.

We reduce a sparse k -SAT instance to a q -DOMSET instance in the following manner:

Let $q = k$. Create an undirected graph G as follows:

1. Create q groups $X_1, \dots, X_q$ of n variables. Create a node X_{i_j} for each possible assignment of a group X_i . That gives us $q2^{n/q}$ groups.
Also create a node n_s for each clause ($1 \leq s \leq m$). Hence we have a total of $q2^{n/q} + m$ nodes.
2. Every X_i is a complete subgraph. If there exists a node X_{i_j} that satisfies some clause n_s create an edge between them.

We claim that G has a dominating set of size q if and only if the k -SAT instance is satisfiable.

($\implies$)

Let S be a dominating set with $|S| = q$. Then S contains exactly one node from each group X_i since for any node $u \in X_i$ whose assignment satisfies no clause, u has no edges to any clause node. Then u 's only neighbors are other nodes in X_i , so u can only be dominated if $S \cap X_i \neq \emptyset$. Since this holds for every group and there are q groups with $|S| = q$, we conclude $|S \cap X_i| = 1$ for each i . (If two nodes in X_i satisfy a clause we just pick one since those nodes are neighbors anyway).

($\impliedby$)

Suppose σ is a satisfying assignment. For each group X_i , let v_i be the node corresponding to σ 's assignment on X_i 's variables, and let $S = \{v_1, \dots, v_q\}$. We claim S is a dominating set of size q :

- Every node $u \in X_i$ is dominated by v_i , since X_i is a complete subgraph and thus v_i is adjacent to every other node in X_i .
- Every clause node n_s is dominated: since σ is satisfying, some group X_i contains a variable that satisfies clause s . Then v_i , which agrees with σ on X_i , also satisfies s , so the edge $v_i \sim n_s$ exists by construction.

Thus we have reduced sparse k -SAT to q -DOMSET. Now we show that if q -DOMSET having an algorithm that solves it in $O(n^{q-\delta})$ time, where n is the number of nodes in the graph, by contradiction. Assume such an algorithm exists. Then,

$$O(n^{q-\delta}) = O\left(\left(q2^{n/q} + m\right)^{q-\delta}\right)$$

By the sparsification lemma we have $m = \Theta(n) = \alpha n$ for some α

$$O(n^{q-\delta}) = O\left(\left(k2^{n/k} + \alpha n\right)^{q-\delta}\right)$$

Note that $2^{n/k}$ clearly dominates, so we drop the other term and the constant k .

$$O(n^{q-\delta}) = O\left(\left(2^{n/k}\right)^{q-\delta}\right) = O\left(2^{n(1-\delta/k)}\right)$$

Since δ/k is positive ($\delta > 0$ and k fixed positive constant), this gives us an $O(2^{n(1-\delta/k)})$ algorithm (which is strictly less than $O(2^n)$) that solves k -SAT, contradicting SETH.

Thus we cannot solve q -DOMSET in $O(n^{q-\delta})$ time.